

COMMISSION ÉLECTROTECHNIQUE INTERNATIONALE

(affiliée à l'Organisation Internationale de Normalisation — ISO)

RECOMMANDATION DE LA CEI

INTERNATIONAL ELECTROTECHNICAL COMMISSION

(affiliated to the International Organization for Standardization — ISO)

IEC RECOMMENDATION

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1971

Schémas, diagrammes, tableaux

Première partie: Définitions et classification

Diagrams, charts, tables

Part 1: Definitions and classification



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

DIAGRAMS, CHARTS, TABLES Part 1: Definitions and classification

FOREWORD

- 1) The formal decisions or agreements of the IEC on technical matters, prepared by Technical Committees on which all the National Committees having a special interest therein are represented, express, as nearly as possible, an international consensus of opinion on the subjects dealt with.
- 2) They have the form of recommendations for international use and they are accepted by the National Committees in that sense.
- 3) In order to promote this international unification, the IEC expresses the wish that all National Committees having as yet no national rules, when preparing such rules, should use the IEC recommendations as the fundamental basis for these rules in so far as national conditions will permit.
- 4) The desirability is recognized of extending international agreement on these matters through an endeavour to harmonize national standardization rules with these recommendations in so far as national conditions will permit. The National Committees pledge their influence towards that end.

PREFACE

This Recommendation has been prepared by IEC Technical Committee No. 3, Graphical Symbols.

Technical Committee No. 3 has established a great number of Recommendations containing graphical symbols for use on diagrams in the field of electrotechnology (see IEC Publication 117, Recommended Graphical Symbols). After having covered most of the needs for graphical symbols, with the exception of those for new fields of electrotechnology still under consideration, Technical Committee No. 3 found it advisable to supplement the symbol publications with Recommendations for the preparation of diagrams.

These Recommendations will include: definitions and classification of diagrams, charts and tables; guiding principles for use and combination of graphical symbols; guiding principles for preparation of diagrams; guiding principles for supplementing or replacing diagrams by charts and tables; item designation, etc.

These Recommendations will be published separately under the general heading of IEC Publication 113, Diagrams, Charts, Tables. This Publication forms Part 1 of this series. It is a revision of Publication 113 (1959), the following modifications having been made:

- 1) Tables will in future replace diagrams and charts to an increasing extent. They have therefore been given more emphasis in the text and are mentioned in the title.
- 2) Some of the alternative terms appearing in Publication 113 have been omitted, e.g. "elementary diagram", "terminal connection diagram" and "one-line representation".
- 3) Some terms have been replaced by new ones, e.g. "schéma de raccordement aux bornes" by "schéma des bornes", "représentation rassemblée" by "représentation assemblée", "représentation semi-développée" by "représentation rangée" and "separated representation" by "detached representation".
- 4) Two concepts have been inserted, i.e. "location diagram or table" and "equivalent circuit diagram".
- 5) A number of definitions have been clarified and some renumbering of clauses has been necessary.

The work on this revision was started in March 1968 and, as a result of the meeting held in London in 1968, a draft was submitted to the National Committees for approval under the Six Months' Rule in March 1969.

The following countries voted explicitly in favour of publication of Part 1:

Australia
Austria
Belgium
Czechoslovakia
Denmark
Finland
France
Germany
Iran
Israel

Italy
Japan
Netherlands
Norway
Poland
Sweden
Switzerland
Turkey
United Kingdom
United States of America

DIAGRAMS, CHARTS, TABLES

Part 1: Definitions and classification

SECTION ONE — GENERAL

1. Scope

The purpose of this Recommendation is to define and to classify diagrams, charts and tables used in electrotechnology in order to explain functions or show connections and to give information for manufacture, installation and maintenance.

Two kinds of classification have been adopted:

- a) classification according to purpose;
- b) classification according to method of representation.

Note. — In practice, the type of diagram is determined by its purpose, the method(s) of representation being chosen for convenience.

2. Definitions

2.1 Diagram

A diagram may show the manner in which the various parts of a network, installation, group of apparatus or items of an apparatus are interrelated and/or interconnected.

2.2 Chart

A chart (U.S.A.: graph) may show the interrelation between:

- a) different operations;
- b) operations and time;
- c) operations and physical quantities;
- d) the states of several items.

2.3 Table

A table replaces or supplements a diagram or a chart.

SECTION TWO — CLASSIFICATION ACCORDING TO PURPOSE

The main classifications are:

- a) explanatory diagrams (Clause 3);
- b) explanatory charts or tables (Clause 4);
- c) wiring diagrams or wiring tables (Clause 5);
- d) location diagrams or tables (Clause 6).

3. Explanatory diagrams

Explanatory diagrams are intended to facilitate the study and understanding of the functioning of an installation or equipment. Three types are defined as follows.

3.1 *Block diagram*

Relatively simple diagram to facilitate the understanding of the principle of operation. It is a diagram in which an installation or equipment together with its functional interrelationships are represented by symbols, block symbols or pictures without necessarily showing all the connections.

3.2 *Circuit diagram (U.S.A.: schematic diagram)*

Explanatory diagram intended to facilitate the understanding of the functioning in detail. It shows by symbols an installation or part of an installation and the electrical connections and other links concerned with its operation.

3.3 *Equivalent circuit diagram*

Special type of circuit diagram for the analysis and calculation of circuit characteristics.

4. **Explanatory charts or tables**

Explanatory charts or tables are intended to facilitate the study of diagrams and to give additional information. Two examples are:

4.1 *Sequence chart or table*

Gives the successive operations in a specified order.

4.2 *Time sequence chart or table*

Is one which in addition takes account of the time intervals between successive operations.

5. **Wiring diagrams or wiring tables**

Wiring diagrams (U.S.A.: connection diagrams) are intended to guide the making and checking of the connections of an installation or equipment. For an equipment, they show the internal or external connections or both. The diagrams may sometimes show the layout of the different parts and accessories such as terminal blocks and the wiring between them.

5.1 *Unit wiring diagram*

Diagram representing all connections within a unit of an installation.

5.2 *Interconnection diagram*

Diagram representing the connections between the different units of an installation.

5.3 *Terminal diagram*

Diagram showing the terminals and the internal and/or external conductors connected to them.

Note. — Any of the wiring diagrams (U.S.A.: connection diagrams) can be replaced or supplemented by a *table*.

6. Location diagrams or tables

A location diagram or table contains detailed information about the location of parts of the equipment, e.g. terminal blocks, plug-in units, sub-assemblies, modules, etc. It shows the item designations used in related diagrams and tables.

Note. — A location diagram need not necessarily be to scale.

Note referring to Clauses 3 to 6. — Several types of diagrams may be combined into a single diagram, forming a *mixed diagram*. The same drawing may form both an explanatory and wiring diagram.

SECTION THREE — CLASSIFICATION ACCORDING TO METHOD OF REPRESENTATION

The method of representation is distinguished by:

- a) the number of conductors, devices or elements represented by a single symbol (Clause 7);
- b) the arrangement of the symbols representing the elements or parts of an item of apparatus (e.g. detached or assembled) (Clause 8);
- c) the placing of the symbols to correspond with the topographical layout of the devices (Clause 9).

7. Number of conductors

According to the number of conductors, devices or elements represented by a single symbol, the following two methods of representation may be distinguished.

7.1 Single-line representation

Two or more conductors are represented by a single line. In particular, a single line may represent:

- circuits of a multiphase system;
- circuits which have a similar electrical function;
- circuits or conductors which belong to the same signal path;
- circuits which follow the same physical route;
- conductor symbols which would follow the same route on the diagram.

Several similar items of apparatus may accordingly be represented by a single symbol.

7.2 Multi-line representation

Each conductor is represented by an individual line.

8. Arrangement of symbols

According to the arrangement of the symbols representing the elements or parts of an item of apparatus on the diagram, there are the following methods of representation.

8.1 Assembled representation

The symbols for the different parts of an item of apparatus or of an installation or equipment are drawn in close proximity on the diagram.

8.2 *Semi-assembled representation*

The symbols for the different parts of an item of apparatus or of an installation are separated and arranged in such a way that the symbols for mechanical linkages between the parts which work together may be drawn easily.

8.3 *Detached representation*

The symbols for the different parts of an item of apparatus or of an installation are separated and arranged in such a way that the circuits may easily be followed.

9. **Topographical representation**

The positions of the symbols on the diagram correspond wholly or partly to the topographical (physical) location of items represented.

The following are examples where topographical representation can be used:

- wiring diagrams;
- architectural diagrams;
- network diagrams.

Note referring to Clauses 7 to 9. — Several of these methods of representation may be used on the same diagram.

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RECOMMANDATION DE LA C. E. I.

INTERNATIONAL ELECTROTECHNICAL COMMISSION

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I. E. C. RECOMMENDATION

Feuille de modification N° 2

Juin 1959

Amendment Sheet No. 2

June 1959

à la Publication 94 (Première édition 1957)

to Publication 94 (First edition-1957)

**Recommandations relatives aux systèmes
d'enregistrement et de lecture sur
bandes magnétiques: Dimensions
et caractéristiques.**

**Recommendations for magnetic tape
recording and reproducing systems:
Dimensions and characteristics.**

La modification ci-après a été discutée par le Comité d'Etudes N° 29, à Stockholm en juillet 1958, et a été approuvée sous la Règle des Six Mois.

The following amendment, which was discussed by Technical Committee No. 29, in Stockholm in July 1958, has been approved under the Six Months' Rule.

Page 6

Chapitre D: Equipement d'enregistrement et de lecture.

Remplacer l'article D 1.2.1 et le nota par ce qui suit:

D 1.2.1 Sens d'enroulement de la bande

Usages professionnels:

Si la face supérieure d'une bobine se distingue par une marque, une étiquette ou une dissymétrie de construction, la bande sera enroulée de telle manière que pendant la reproduction elle puisse se dérouler en sens inverse des aiguilles d'une montre.

Usages amateurs:

Aucune recommandation.

Page 7

Section D: Recording and reproducing equipment.

Replace Clause D 1.2.1 and the note by the following:

D 1.2.1 Direction of tape winding

For professional use:

If the top surface of a spool is distinguished by markings, by a label or because of an asymmetrical construction, then the tape shall be wound in such a way that during reproduction it may be unwound in an anti-clockwise direction.

For domestic use:

No recommendation is given.

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